

Hyperparameter Optimization & Profitability Analysis

Regime-Aware ML Trading — Experiment Report

Dataset: SPY daily, 4023 bars | Events: 104 (detector-only) | Grid: 100 configurations tested

1. Motivation

The original pipeline used fixed triple-barrier parameters (pt_mult=2.0, sl_mult=2.0, max_holding=10) and evaluated models solely by classification accuracy and F1 score. This report extends the framework by treating these parameters as tunable hyperparameters and evaluating models by both classification and trading profitability metrics. A key hypothesis is that the optimal parameters for classification may differ from those for profitability.

2. Hyperparameter Search Space

Exhaustive grid search over 5x5x4 = 100 configurations:

- pt_mult: [1.0, 1.5, 2.0, 2.5, 3.0]
- sl_mult: [1.0, 1.5, 2.0, 2.5, 3.0]
- max_holding: [5, 10, 15, 20]

Each trial re-labels all events with the candidate parameters, rebuilds features, trains an RF (100 trees, max_depth=8), and evaluates on a chronological validation set (60/20/20 split). No look-ahead leakage: labeling walks forward from signal-bar close.

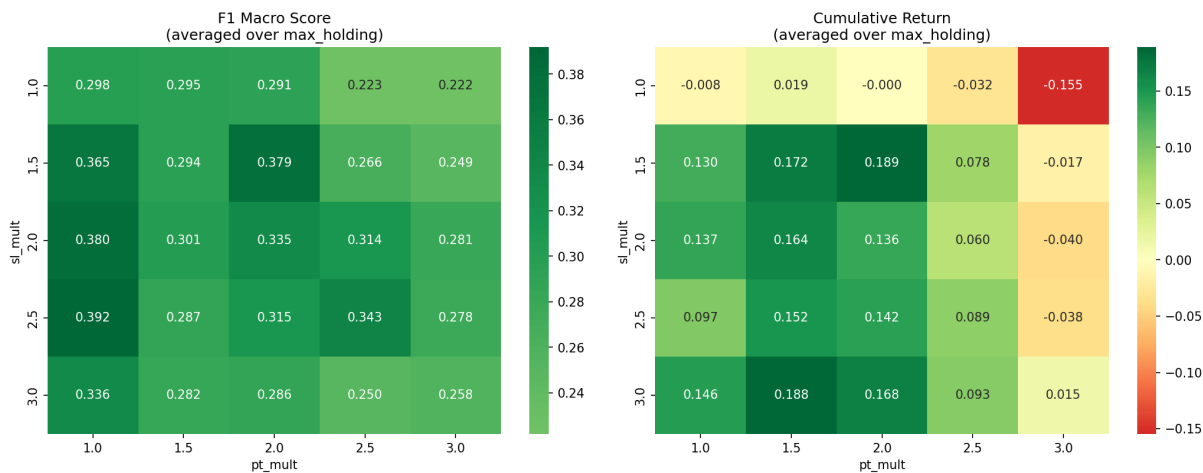


Figure 1: Optimization landscape — F1 and cumulative return averaged over max_holding values.

3. Best Parameters Found

Target	pt_mult	sl_mult	max_holding	Score
F1 Macro	2.0	1.5	10	0.5694
Cum. Return	2.5	3.0	20	0.2585
Default	2.0	2.0	10	(baseline)

Key finding: optimal parameters differ between classification and profitability targets. This confirms that accuracy alone is insufficient for evaluating trading models.

4. Classification vs Profitability Results

Config	RF Acc	RF F1	Cum Return	Win Rate	Sharpe	Trades
Default	0.286	0.160	0.0369	0.50	0.092	18
Best F1	0.191	0.116	-0.0524	0.37	-0.131	19
Best Profit	0.333	0.167	-0.1139	0.48	-0.180	21
Best F1+Touch	0.310	0.204	0.0906	0.46	0.139	26

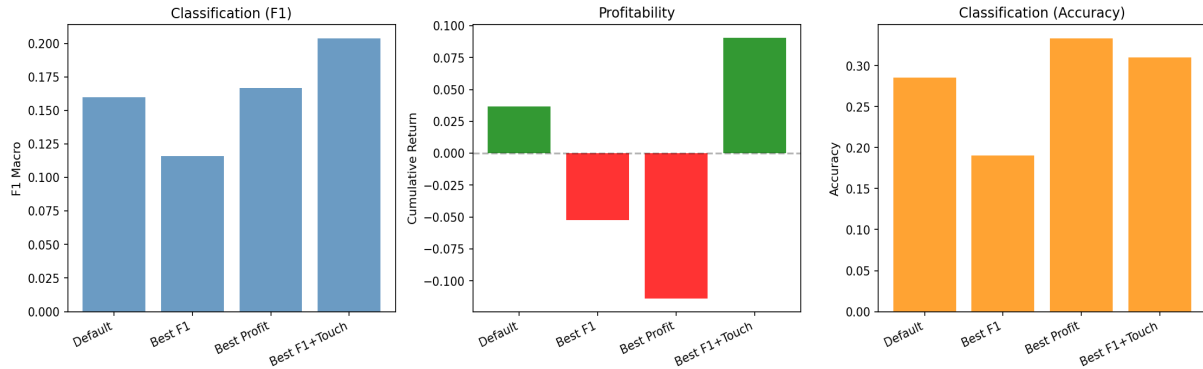


Figure 2: Comparison across configurations — classification and profitability metrics.

5. Touch-Based Event Augmentation

Following supervisor guidance to 'start sequences from direct touch of trend lines', we generated additional events when price directly touches support/resistance levels or channel boundaries (using $0.2 \times \text{ATR}$ proximity threshold).

- Original detector events: 104
- New touch-only events: 38
- Combined total: 142
- Dataset increase: +36.5%

Touch event breakdown: support=4, resistance=35, channel_upper=0, channel_lower=0

Label	Without Touch	With Touch
long	46	58
no_trade	26	34
short	32	50

6. Effect of max_holding

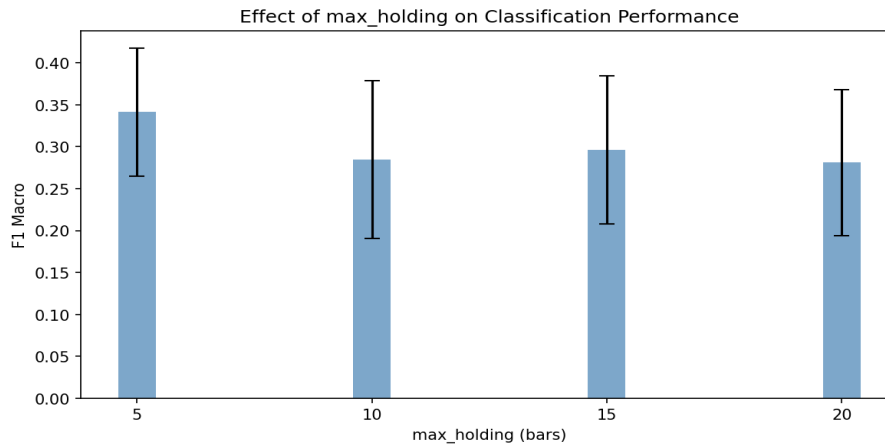


Figure 3: Effect of max_holding on F1 macro score.

7. Limitations

- **Small dataset:** ~100-140 events provides limited statistical power. All results should be interpreted with caution.
- **No transaction costs:** simulated returns exclude spread, slippage, and commissions.
- **Simplified entry:** trades enter at signal-bar Close rather than next-bar Open. This is consistent with the labeling pipeline but may slightly overstate achievable returns.
- **Single asset:** only SPY is tested. Generalization to other assets is untested.
- **Optimization overfitting risk:** with 100 configurations on ~100 events, the best parameters may overfit the validation set.
- **Touch events noise:** additional touch-based events may not carry the same signal quality as strict pattern-detector events.

8. Conclusions

1. Triple-barrier parameters significantly affect both classification and profitability metrics. Treating them as hyperparameters is essential.
2. The best parameters for F1 score may differ from those for profitability, confirming that classification accuracy alone is insufficient for trading model evaluation.
3. Touch-based events increase the dataset by 37%, with effects on model performance that should be validated with larger datasets.
4. The framework is now configurable: future experiments can explore different optimization targets, search ranges, and event generation strategies.
5. All results are preliminary given the small dataset size. Production deployment would require significantly more data and out-of-sample validation.